USER INTERFACE DESIGN EXP:2 MEMORY TEST USING CHUNKING

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Aim

To design and evaluate a user interface that demonstrates how grouping (chunking) visual elements such as logos improves short-term memory recall.

Objective

- To understand the concept of chunking in Human–Computer Interaction.
- To design memory recall UI screens using Figma.
- To analyse how categorized (grouped) logos improve recall accuracy.
- To measure user performance through a result screen.

Tools Used

- Figma (UI Design & Prototyping)
- Desktop/Laptop

Theory

Chunking is a cognitive principle in which individual pieces of information are grouped into meaningful units. According to cognitive psychology, short-term memory can hold limited items (7±2 items). When information is grouped into categories, it reduces cognitive load and increases recall efficiency.

In UI/UX design, chunking is used to:

- Organize content clearly
- Improve usability

- Enhance recognition and recall
- Reduce memory overload

This experiment demonstrates how categorized logos improve memory performance compared to random presentation.

Experimental Design

The experiment consists of multiple UI frames designed in Figma:

1. Home Screen (Instruction Page)
2. Memory Display Phase (Chunking Phase)
3. Recall Phase
4. Result Screen

A. Home Screen (Instruction Page)

Description

This screen introduces the experiment and provides instructions to the user.

Contents

- Title: **Memory Recall Task**
- Instruction box explaining the task
- START button to begin the experiment

Instructions Displayed

- Users will be shown groups of logos.
- Viewing time is limited.
- Try to remember as many logos as possible.
- After viewing, users must recall them in the next screen.

This screen contains a centered instruction card and a black START button at the bottom.

B. Memory Display Phase (Chunking Phase)

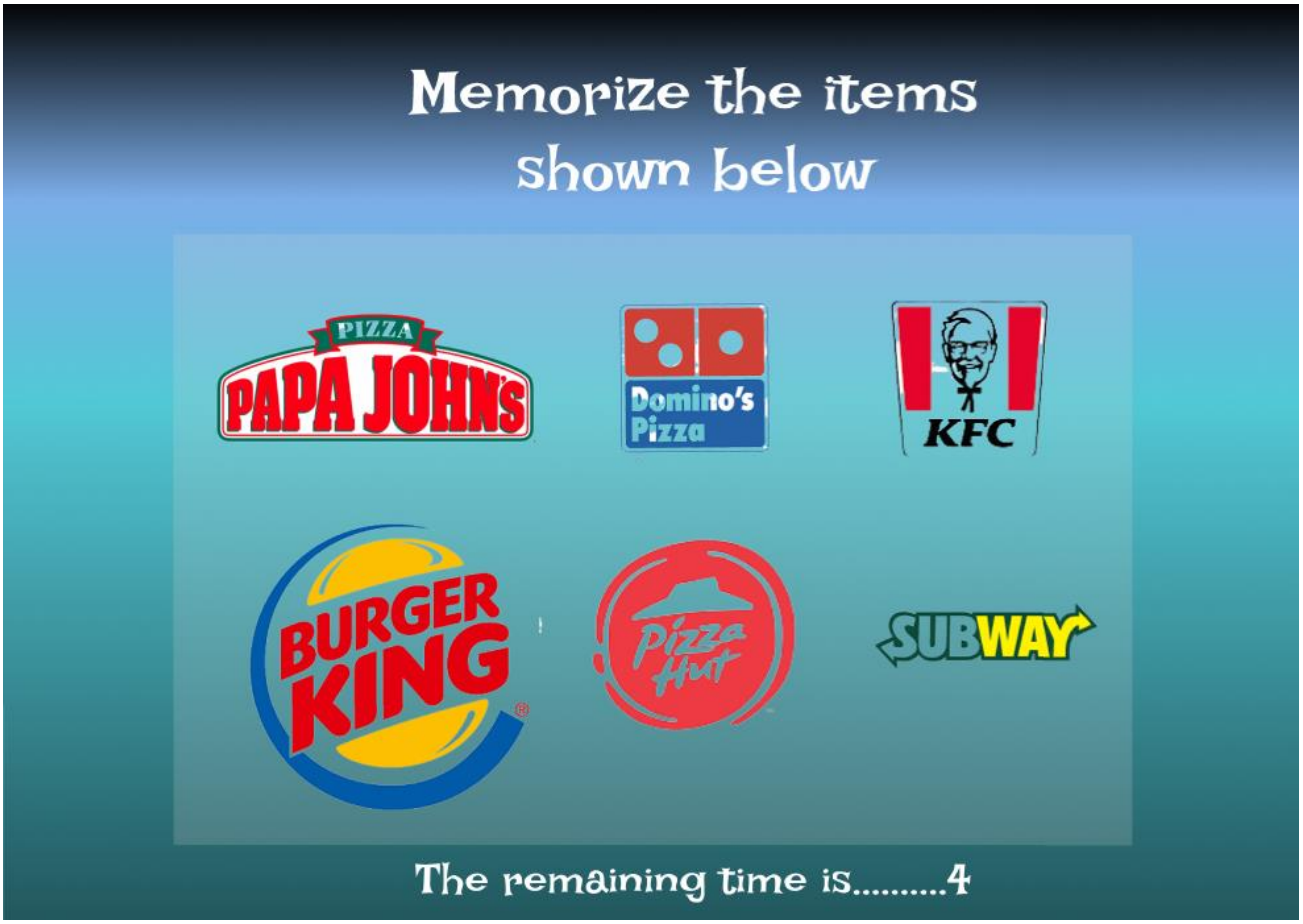
Description

In this phase, food brand logos are displayed as a single semantic group. By presenting items that belong to the same category, the system applies chunking to help users process and retain information more effectively during the memorization period.

Key Design Features

- Clear visual spacing between the logos
- Uniform alignment and grid structure
- Clean background with focus on logos
- Time-based transition (simulated in prototype)

This structured layout helps users mentally organize information.



C. Recall Phase

Description:

After viewing the grouped logos, users move to the recall screen.

Recall Method Used

- Multiple random logos are displayed
- Users select logos they remember seeing

UI Elements

- Categories displayed again
- Progress bar showing completion level
- DONE button
- CLEAR button

The progress bar visually indicates how many items the user has selected.

This phase measures how many items the user can correctly recall.

D. Result Screen

Description

This screen displays the user’s performance after submission.

Information Shown

- Score indicator (progress bar)
- Number of correctly recalled items

Example:

"You recalled 80% correctly!!"

The result screen includes:

- A black progress bar for correct recall
- Retry or Exit buttons

Observation

- Users were able to recall more logos when they were grouped into categories.
- Categorized presentation improved recognition speed.
- Visual grouping reduced memory overload.
- The progress bar encouraged active participation

Result

The experiment shows that chunked (grouped) visual information significantly improves short-term memory recall compared to unstructured or random presentation.

Users demonstrated higher recall accuracy when logos were categorized.

Conclusion

Chunking is an effective UI design strategy in Human–Computer Interaction.

By grouping related visual elements such as brand logos:

- Cognitive load is reduced
- Memory retention improves
- User performance increases
- Interface usability is enhanced

This experiment successfully demonstrates the practical application of cognitive psychology principles in interface design.